



# Moove Education Platform

## Software Requirements Specification (SRS)

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Moove

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## 0.1 Introduction

### 0.1.1 Purpose

This document defines the functional and non-functional requirements for the Moove Education Platform, an Open edX-based learning management system deployed using Tutor. It serves as the living source of truth for system capabilities, guiding continuous development, internal quality assurance, and security assessment activities. The requirements are aligned with Ethiopian regulatory frameworks, including the Critical Mass Cyber Security Requirement Standard (CMCSRS), the National Cybersecurity Policy, the Computer Crime Proclamation, and the Electronic Signature Proclamation.

### 0.1.2 Scope

The system provides a comprehensive online learning environment supporting:

- Course creation, authoring, and management (Educator features).
- Learner enrollment, progression, and engagement (Learner features).
- Content delivery (video, text, interactive components).
- Assessments (quizzes, assignments, peer reviews, proctored exams).
- Collaboration (discussions, teams, wikis).
- Analytics and reporting.
- Administrative controls.

### 0.1.3 System Overview and Actors

The platform is a client-server application based on Open edX, deployed in Docker containers orchestrated by Tutor. It uses a microservices architecture with separate containers for LMS, CMS, MFE, databases, and supporting services.

#### Actors

- **Educator (Instructor/Staff):** Creates and manages courses, creates content, assesses learners, moderates discussions, views analytics.
- **Learner (Student):** Enrolls in courses, accesses content, completes assessments, participates in discussions, tracks progress.
- **Administrator:** Manages users, courses, site configuration, and system health.
- **System:** Automated processes (notifications, email, background tasks).

### 0.1.4 System Constraints

- The system must comply with Ethiopian laws and regulations, including the Computer Crime Proclamation and the Electronic Signature Proclamation.
- All user data must be stored and processed within the country's jurisdiction.
- The system must support English and Amharic (planned) interfaces.
- The system shall be accessible via modern web browsers and mobile devices.

### 0.1.5 Assumptions and Dependencies

- The platform assumes reliable internet connectivity for both learners and educators.
- Third-party services (e.g., video hosting, proctoring) are available and secured.
- The underlying Tutor deployment and container infrastructure are maintained and updated regularly.

## 0.2 Functional Requirements – Educator Features

### 0.2.1 Course Creation & Planning Module

**General Description** Provides tools for educators to create, plan, and structure courses.

#### Requirement UFR-OE-001-01: Create a New Course

1. **Title:** Create a new course.
2. **UML Traceability:** Use Case: “Create Course”; Activity: “Course Creation Flow”.
3. **Actor:** Educator (Instructor/Staff).
4. **Pre-conditions:**
  - Educator is logged in with course creation permissions.
  - The organisation code, course number, and run combination is unique.
5. **Post-conditions:**
  - A new course skeleton is created in the CMS.
  - The course appears in the educator's course list.
  - An audit log entry is recorded.
6. **Normal Flow:**
  - (a) Educator navigates to Studio (CMS).
  - (b) Clicks “New Course”.
  - (c) Enters course title, organisation code, course number, and course run.

- (d) Sets start date, enrollment start/end dates, and course visibility.
- (e) Clicks “Create”.
- (f) System validates uniqueness and creates the course ID.

#### 7. Alternative Flows:

- Alt 1: Duplicate course ID → validation error.

#### 8. Business Rules: Course ID format: org+course+run; must be unique.

#### 9. Input/Output: Input: title, org, number, run, start date, enrollment dates. Output: Course object with ID.

#### 10. Acceptance Criteria:

```
Given I am an educator
When I create a course with title "Introduction to Physics"
Then I receive a 201 Created
And the course appears in my course list
```

### Requirement UFR-OE-001-02: Plan a New Course (Outline)

1. **Title:** Plan course outline.
2. **UML Traceability:** Activity: “Course Outline Flow”.
3. **Actor:** Educator.
4. **Pre:** Course exists.
5. **Post:** Course outline (sections, subsections, units) is defined.
6. **Normal Flow:**
  - (a) Educator opens the course outline.
  - (b) Clicks “Add Section” and enters a name.
  - (c) Within a section, clicks “Add Subsection” and enters a name.
  - (d) Within a subsection, clicks “Add Unit” to create content pages.
  - (e) Drags and drops to reorder.
7. **Input/Output:** Input: section/subsection/unit metadata. Output: updated course structure.

### Requirement UFR-OE-001-03: Enroll Learners in a Course

1. **Title:** Enroll learners manually.
2. **UML Traceability:** Use Case: “Manage Enrollments”.
3. **Actor:** Educator.
4. **Pre:** Course exists and enrollment is open.
5. **Post:** Learner is enrolled.
6. **Normal Flow:**
  - (a) Educator goes to “Instructor Dashboard” → “Enrollment”.
  - (b) Enters learner email(s) and selects enrollment track (audit/verified).
  - (c) Clicks “Enroll”.
  - (d) System sends confirmation email.
7. **Alt:** Bulk enrollment via CSV upload.
8. **Acceptance:** Bulk upload processes 500+ enrollments within 5 minutes.

## 0.2.2 Instructional Design Concepts

### Requirement UFR-OE-002-01: Apply Instructional Design Principles

1. **Title:** Apply instructional design concepts.
2. **UML Traceability:** – (guideline).
3. **Actor:** Educator.
4. **Pre:** Course exists.
5. **Post:** Course is structured using sound pedagogical design.
6. **Normal Flow:**
  - (a) Educator defines learning objectives for each section.
  - (b) Creates assessments aligned with objectives.
  - (c) Uses Open edX best practices (backward design, Bloom’s taxonomy).

## 0.2.3 Learner Engagement & Communication Module

### Requirement UFR-OE-003-01: Course Discussions

1. **Title:** Configure course discussions.
2. **UML Traceability:** Use Case: “Manage Discussions”.
3. **Actor:** Educator.

4. **Pre:** Course exists.
5. **Post:** Discussion forums enabled.
6. **Normal Flow:**
  - (a) Educator goes to “Advanced Settings” and enables discussions.
  - (b) Configures topics and moderation policies.

#### Requirement UFR-OE-003-02: Bulk Emails

1. **Title:** Send bulk emails to learners.
2. **UML Traceability:** Sequence: “Bulk Email Flow”.
3. **Actor:** Educator.
4. **Pre:** Course has enrolled learners.
5. **Post:** Email sent asynchronously.
6. **Normal Flow:**
  - (a) Educator composes email, selects recipients (all learners, specific segments).
  - (b) Clicks “Send”.
  - (c) System queues via Celery.

#### Requirement UFR-OE-003-03: Course Updates

1. **Title:** Publish course updates.
2. **UML Traceability:** – (simple notification).
3. **Actor:** Educator.
4. **Post:** Update appears on course home page.
5. **Normal Flow:** Educator creates an update post (title, content) and publishes.

### 0.2.4 Content Creation & Management Module

#### Requirement UFR-OE-004-01: Create and Manage Components

1. **Title:** Add and manage course components.
2. **UML Traceability:** Activity: “Component Management”.
3. **Actor:** Educator.
4. **Pre:** Unit exists.
5. **Post:** Component added to unit.
6. **Normal Flow:**



- (a) Educator selects a unit.
- (b) Clicks “Add New Component” and chooses type (HTML, Video, Problem, Discussion, etc.).
- (c) Configures component content.
- (d) Saves.

#### Requirement UFR-OE-004-02: Manage Text Components

1. **Title:** Create and edit HTML/text components.
2. **Actor:** Educator.
3. **Normal Flow:** Uses rich text editor or raw HTML; supports LaTeX, code snippets.

#### Requirement UFR-OE-004-03: Manage Video Components

1. **Title:** Upload and manage video content.
2. **UML Traceability:** Use Case: “Manage Video”.
3. **Actor:** Educator.
4. **Pre:** Video file available.
5. **Post:** Video player embedded.
6. **Normal Flow:** Uploads video file or enters URL; system processes and displays player with transcripts, speed control.

#### Requirement UFR-OE-004-04: Control Content Visibility & Access

1. **Title:** Control content visibility and access.
2. **UML Traceability:** – (settings).
3. **Actor:** Educator.
4. **Normal Flow:** Sets release dates, prerequisites, and visibility rules.

### 0.2.5 Components & Activities Module

#### Requirement UFR-OE-005-01: Dropdown Problem

1. **Title:** Create dropdown problem.
2. **UML Traceability:** – (component type).
3. **Actor:** Educator.
4. **Pre:** Unit exists.
5. **Post:** Problem added to unit.
6. **Normal Flow:**

- (a) Adds a Problem component.
- (b) Selects “Dropdown” type.
- (c) Enters question and options.
- (d) Marks correct answer.
- (e) Saves.

#### Requirement UFR-OE-005-02: Multi-Select Problem

1. **Title:** Create multi-select problem.
2. **Actor:** Educator.
3. **Normal Flow:** Adds Problem component, selects “Multi-Select”, enters options and marks multiple correct answers.

#### Requirement UFR-OE-005-03: Open Response Assessments (ORA)

1. **Title:** Create open response assessment.
2. **UML Traceability:** Activity: “ORA Flow”.
3. **Actor:** Educator.
4. **Normal Flow:**
  - (a) Adds Advanced component → “Open Response Assessment”.
  - (b) Defines prompt and rubric.
  - (c) Configures assessment steps (self, peer, staff).
  - (d) Sets deadlines.
  - (e) Saves.

#### Requirement UFR-OE-005-04: Single Select Problem

1. **Title:** Create single select (radio button) problem.
2. **Actor:** Educator.
3. **Normal Flow:** Adds Problem, selects “Single Select”, enters options, marks correct answer.

#### Requirement UFR-OE-005-05: Text Input Problem

1. **Title:** Create text input problem.
2. **Actor:** Educator.
3. **Normal Flow:** Adds Problem, selects “Text Input”, enters question and expected answer(s).

## 0.2.6 Data & Analytics Module

### Requirement UFR-OE-006-01: View Course Enrollments

1. **Title:** View course enrollment data.
2. **UML Traceability:** Use Case: “View Analytics”.
3. **Actor:** Educator.
4. **Normal Flow:** Accesses Instructor Dashboard → Data Downloads → Enrollment Data; views totals, trends, demographics.

### Requirement UFR-OE-006-02: View Learner Data

1. **Title:** View learner data and activity.
2. **Actor:** Educator.
3. **Normal Flow:** Instructor Dashboard → Student Admin; searches learner; views progress, engagement, activity history.

### Requirement UFR-OE-006-03: View Learner Grades

1. **Title:** View and manage learner grades.
2. **UML Traceability:** Sequence: “Grade View”.
3. **Actor:** Educator.
4. **Normal Flow:** Accesses Gradebook; views grades per assignment; adjusts manually if needed; exports CSV.

### Requirement UFR-OE-006-04: View Answer Data

1. **Title:** View answer distribution data.
2. **Actor:** Educator.
3. **Normal Flow:** Data Downloads → Answer Distribution; sees % correct per problem.

### Requirement UFR-OE-006-05: View Certificate Data

1. **Title:** View certificate issuance data.
2. **Actor:** Educator.
3. **Normal Flow:** Data Downloads → Certificate Data; views list of issued certificates and download stats.

## 0.2.7 Accessibility Module

### Requirement UFR-OE-007-01: Open edX Accessibility Guidelines

1. **Title:** Ensure accessibility compliance.
2. **Actor:** Educator.
3. **Description:** Course content must follow Open edX accessibility guidelines (WCAG 2.1 AA).

### Requirement UFR-OE-007-02: Mobile Experience Design

1. **Title:** Design for mobile experience.
2. **Actor:** Educator.
3. **Description:** Course content must be responsive and functional on mobile devices.

### Requirement UFR-OE-007-03: Accessibility Best Practices Checklist

1. **Title:** Follow accessibility best practices.
2. **Actor:** Educator.
3. **Description:** Use alt text, proper heading structure, accessible color contrast, keyboard navigation.

### Requirement UFR-OE-007-04: Manage Video Transcripts

1. **Title:** Upload and manage video transcripts.
2. **Actor:** Educator.
3. **Normal Flow:** Uploads .srt/.vtt files for video components; transcripts appear as captions.

## 0.2.8 Import/Export / OLX Module

### Requirement UFR-OE-008-01: Open Learning XML (OLX) Support

1. **Title:** Support for Open Learning XML.
2. **Actor:** Educator.
3. **Description:** Course content can be exported and imported using OLX format.

### Requirement UFR-OE-008-02: Getting Started with OLX

1. **Title:** Access OLX documentation and tools.
2. **Actor:** Educator.
3. **Description:** Provide resources for educators to learn and use OLX.

## 0.3 Functional Requirements – Learner Features

### 0.3.1 Dashboard & Profile Module

#### Requirement UFR-OE-101-01: Explore Dashboard and Profile

1. **Title:** Access and explore learner dashboard.
2. **UML Traceability:** Use Case: “View Dashboard”.
3. **Actor:** Learner.
4. **Pre:** Learner logged in.
5. **Post:** Dashboard displays enrolled courses, progress, announcements.
6. **Normal Flow:**
  - (a) Learner logs in.
  - (b) Dashboard shows course cards with progress bars, recent activity, and site announcements.

#### Requirement UFR-OE-101-02: Access Courses from Dashboard

1. **Title:** Access enrolled courses from dashboard.
2. **Actor:** Learner.
3. **Normal Flow:** Clicks on course card; enters the course.

#### Requirement UFR-OE-101-03: Share Courses on Social Media

1. **Title:** Share course links on social media.
2. **Actor:** Learner.
3. **Normal Flow:** Clicks “Share” button; selects platform (Facebook, Twitter, LinkedIn); link posted.

#### Requirement UFR-OE-101-04: Update Profile Image

1. **Title:** Upload or update profile image.
2. **Actor:** Learner.
3. **Normal Flow:** Clicks on profile, selects image file; system resizes and stores.

#### Requirement UFR-OE-101-05: Update Profile Details

1. **Title:** Update profile information.
2. **Actor:** Learner.
3. **Fields:** Name, email, location, bio, language preferences.
4. **Normal Flow:** Edits fields and saves.

**Requirement UFR-OE-101-06: Add Social Media Links**

1. **Title:** Add personal social media links.
2. **Actor:** Learner.
3. **Normal Flow:** Adds URLs for Twitter, LinkedIn, GitHub, etc.

**Requirement UFR-OE-101-07: Toggle Profile Visibility**

1. **Title:** Control profile visibility.
2. **Actor:** Learner.
3. **Normal Flow:** Toggles visibility of profile sections (public/private).

**Requirement UFR-OE-101-08: View Another Learner's Profile**

1. **Title:** View other learners' profiles.
2. **Actor:** Learner.
3. **Normal Flow:** Clicks on username in discussions or roster; profile opens (public info only).

**Requirement UFR-OE-101-09: Update Course-Specific Settings**

1. **Title:** Update course-specific settings.
2. **Actor:** Learner.
3. **Normal Flow:** In course, clicks avatar → Course Settings; adjusts email notifications, language.

## 0.3.2 Course Navigation Module

**Requirement UFR-OE-102-01: View Course Sections from Navigation Sidebar**

1. **Title:** Navigate course using sidebar.
2. **Actor:** Learner.
3. **Normal Flow:** Sidebar displays all sections/subsections; click to navigate.

## 0.3.3 Enrollment Module

**Requirement UFR-OE-103-01: Enroll in a Course**

1. **Title:** Enroll in a course.
2. **UML Traceability:** Sequence: "Enrollment Flow".
3. **Actor:** Learner.
4. **Pre:** Course open for enrollment.

**5. Post:** Learner enrolled.

**6. Normal Flow:**

- (a) Browses course catalog.
- (b) Clicks “Enroll Now”.
- (c) Selects enrollment track (Audit/Verified).
- (d) Confirms enrollment.

### Requirement UFR-OE-103-02: Open and Start a Course

- 1. Title:** Open and start a course.
- 2. Actor:** Learner.
- 3. Normal Flow:** Clicks “Start Course” or “Resume Course” from dashboard; goes to last visited content.

## 0.3.4 Course Prerequisites Module

### Requirement UFR-OE-104-01: Overview of Prerequisites

- 1. Title:** View course prerequisites.
- 2. Actor:** Learner.
- 3. Normal Flow:** Prerequisites displayed on course info page; learner can check completion status.

### Requirement UFR-OE-104-02: Complete Prerequisite Course

- 1. Title:** Complete prerequisite course.
- 2. Actor:** Learner.
- 3. Normal Flow:** Must complete prerequisite course before enrolling in advanced course.

### Requirement UFR-OE-104-03: Pass Entrance Exam

- 1. Title:** Pass entrance exam.
- 2. Actor:** Learner.
- 3. Normal Flow:** Takes and passes an entrance exam to gain access.

### 0.3.5 Course Content Access Module

#### Requirement UFR-OE-105-01: Troubleshoot Accessing Course Content

1. **Title:** Troubleshoot content access issues.
2. **Actor:** Learner.
3. **Normal Flow:** System provides error messages and support links; contact course staff.

#### Requirement UFR-OE-105-02: Access an Archived Course

1. **Title:** Access archived course.
2. **Actor:** Learner.
3. **Normal Flow:** Views read-only archived course content.

#### Requirement UFR-OE-105-03: View Final Course Grade

1. **Title:** View final course grade.
2. **Actor:** Learner.
3. **Post:** Sees final grade and passing status.

#### Requirement UFR-OE-105-04: View Certificate Status

1. **Title:** View certificate status.
2. **Actor:** Learner.
3. **Normal Flow:** In profile → Certificates, sees status (Eligible, Available, Not Eligible).

### 0.3.6 Course Progress Module

#### Requirement UFR-OE-106-01: Find Course Start Date

1. **Title:** View course start date.
2. **Actor:** Learner.
3. **Normal Flow:** Start date displayed on course info and dashboard.

#### Requirement UFR-OE-106-02: Start or Resume a Course

1. **Title:** Start or resume a course.
2. **Actor:** Learner.
3. **Normal Flow:** Clicks “Start” or “Resume” from dashboard; goes to last visited content.



**Requirement UFR-OE-106-03: Progress Indicators**

1. **Title:** View progress indicators.
2. **Actor:** Learner.
3. **Description:** Visual indicators show completion percentage for sections, subsections, and overall.

**Requirement UFR-OE-106-04: The Progress Page**

1. **Title:** Access the Progress page.
2. **Actor:** Learner.
3. **Normal Flow:** Navigates to Progress page to view detailed completion status.

### 0.3.7 Certificates & Badges Module

**Requirement UFR-OE-107-01: Accessing a Certificate**

1. **Title:** Access earned certificate.
2. **Actor:** Learner.
3. **Post:** Downloads or prints certificate.

**Requirement UFR-OE-107-02: Viewing Earned Badges in Profile**

1. **Title:** View earned badges in profile.
2. **Actor:** Learner.
3. **Normal Flow:** Badges appear in profile; click to view details.

**Requirement UFR-OE-107-03: Sharing Badges on Mozilla Backpack**

1. **Title:** Share badges to Mozilla Backpack.
2. **Actor:** Learner.
3. **Normal Flow:** Clicks “Share to Backpack” and authorises.

### 0.3.8 Video Player Module

**Requirement UFR-OE-108-01: Watching Videos on the Open edX Video Player**

1. **Title:** Use the video player.
2. **Actor:** Learner.
3. **Features:** Play, pause, volume, speed control, full-screen, CC/subtitles.

### 0.3.9 Discussions Module

#### Requirement UFR-OE-109-01: Getting Started with Discussions

1. **Title:** Access course discussions.
2. **Actor:** Learner.
3. **Normal Flow:** Clicks “Discussions” tab; sees threads and topics.

#### Requirement UFR-OE-109-02: Finding and Following Conversations

1. **Title:** Find and follow conversations.
2. **Actor:** Learner.
3. **Features:** Search, filter, follow threads, get notifications.

#### Requirement UFR-OE-109-03: Taking Part in Discussions

1. **Title:** Participate in discussions.
2. **Actor:** Learner.
3. **Normal Flow:** Creates new threads, replies to existing ones.

#### Requirement UFR-OE-109-04: Staying Updated with Notifications

1. **Title:** Receive discussion notifications.
2. **Actor:** Learner.
3. **Normal Flow:** System sends email notifications for followed threads.

### 0.3.10 Assignments & Assessments Module

#### Requirement UFR-OE-110-01: Completing Mathematical and Scientific Assignments

1. **Title:** Complete mathematical assignments.
2. **Actor:** Learner.
3. **Features:** Support for LaTeX, math symbols, scientific notation.

#### Requirement UFR-OE-110-02: Taking a Timed Exam

1. **Title:** Take timed exam.
2. **Actor:** Learner.
3. **Normal Flow:** Starts exam; timer counts down; auto-submit at expiration.

**Requirement UFR-OE-110-03: Proctored Exams**

1. **Title:** Take proctored exam.
2. **Actor:** Learner.
3. **Features:** Proctoring software integration, identity verification.

**Requirement UFR-OE-110-04: Explaining Multiple Choice Answers**

1. **Title:** Provide explanation for multiple choice answers.
2. **Actor:** Learner.
3. **Normal Flow:** Selects answer and optionally enters explanation.

**Requirement UFR-OE-110-05: Completing Essay Assignments**

1. **Title:** Complete essay assignment.
2. **Actor:** Learner.
3. **Normal Flow:** Writes essay in text editor (supports formatting).

**Requirement UFR-OE-110-06: Open Response Assessment Steps**

1. **Title:** Follow ORA workflow.
2. **Actor:** Learner.
3. **Steps:** Submit response, self-assessment, peer assessment, receive score.

**Requirement UFR-OE-110-07: How Grading Is Done In Open Response Assessments**

1. **Title:** Understand ORA grading.
2. **Actor:** Learner.
3. **Description:** System displays grading rubric and peer review scores.

**Requirement UFR-OE-110-08: Completing an Open Response Assessment**

1. **Title:** Complete ORA.
2. **Actor:** Learner.
3. **Normal Flow:** Submits response, performs self/peer assessments.

**Requirement UFR-OE-110-09: Receive Score and Provide Feedback**

1. **Title:** Receive ORA score and provide feedback.
2. **Actor:** Learner.
3. **Normal Flow:** Views score and provides feedback on assessment experience.

**Requirement UFR-OE-110-10: How Peer Assessment Scores Are Calculated**

1. **Title:** Understand peer assessment scoring.
2. **Actor:** Learner.
3. **Description:** System explains scoring algorithm (average, median, adjusted).

**Requirement UFR-OE-110-11: Canceled Responses**

1. **Title:** Handle canceled responses.
2. **Actor:** Learner.
3. **Normal Flow:** Can cancel submission before assessment.

**0.3.11 Team Projects Module****Requirement UFR-OE-111-01: About Teams and Topics**

1. **Title:** Understand teams and topics.
2. **Actor:** Learner.
3. **Description:** System provides team and topic overview.

**Requirement UFR-OE-111-02: Search for a Team**

1. **Title:** Search for a team.
2. **Actor:** Learner.
3. **Normal Flow:** Searches teams by name, topic, or member.

**Requirement UFR-OE-111-03: Join a Team**

1. **Title:** Join an existing team.
2. **Actor:** Learner.
3. **Post:** Added to team.

**Requirement UFR-OE-111-04: Leave a Team**

1. **Title:** Leave a team.
2. **Actor:** Learner.
3. **Post:** Removed from team.

**Requirement UFR-OE-111-05: Create a Team**

1. **Title:** Create a new team.
2. **Actor:** Learner.
3. **Post:** Team created; learner becomes member.

**Requirement UFR-OE-111-06: Participating in Team Discussions**

1. **Title:** Participate in team discussions.
2. **Actor:** Learner.
3. **Normal Flow:** Team members discuss within team-specific forum.

**0.3.12 Search Module****Requirement UFR-OE-112-01: Overview of Search**

1. **Title:** Understand search capabilities.
2. **Actor:** Learner.
3. **Description:** Full-text search across course content.

**Requirement UFR-OE-112-02: Search a Single Course**

1. **Title:** Search within a course.
2. **Actor:** Learner.
3. **Normal Flow:** Enters query; returns relevant content within course.

**Requirement UFR-OE-112-03: Search All Your Courses**

1. **Title:** Search across all courses.
2. **Actor:** Learner.
3. **Normal Flow:** Enters query; returns results from all enrolled courses.

**0.3.13 Bookmarking Module****Requirement UFR-OE-113-01: Add or Remove a Bookmark**

1. **Title:** Bookmark course content.
2. **Actor:** Learner.
3. **Normal Flow:** Clicks bookmark icon to save/remove bookmark.

**Requirement UFR-OE-113-02: View Your Bookmarks**

1. **Title:** View bookmarked content.
2. **Actor:** Learner.
3. **Normal Flow:** Accesses bookmarks page to view and navigate saved items.

### 0.3.14 Notifications & Preferences Module

#### Requirement UFR-OE-114-01: Where Notifications Appear

1. **Title:** Access notifications.
2. **Actor:** Learner.
3. **Normal Flow:** Notifications appear in dashboard, email, and in-app alerts.

#### Requirement UFR-OE-114-02: Activity Types

1. **Title:** Understand notification activity types.
2. **Actor:** Learner.
3. **Description:** Types include discussion replies, assignment due dates, course updates.

#### Requirement UFR-OE-114-03: Managing Preferences

1. **Title:** Manage notification preferences.
2. **Actor:** Learner.
3. **Normal Flow:** Configures which notifications to receive and via which channels.

#### Requirement UFR-OE-114-04: Notifications Expiry

1. **Title:** Notification expiry.
2. **Actor:** System.
3. **Description:** Notifications expire after 30 days.

### 0.3.15 Course Wiki Module

#### Requirement UFR-OE-115-01: Using the Course Wiki

1. **Title:** Use the course wiki.
2. **Actor:** Learner.
3. **Normal Flow:** Accesses wiki from course navigation; views and edits (if allowed).

#### Requirement UFR-OE-115-02: View Live Course

1. **Title:** View live course content.
2. **Actor:** Learner.
3. **Normal Flow:** Accesses content that is currently active/published.

## 0.4 Security and Compliance Requirements (Non-Functional)

This section defines mandatory security requirements derived from INSA's CMCSRS, the Computer Crime Proclamation, and the Electronic Signature Proclamation. All requirements are traceable to the *Security Architecture & Threat Model* document.

### 0.4.1 Access Control (CMCSRS 6.17)

- SEC-NFR-001-1. The system must enforce role-based access control (RBAC) for all user actions.
- SEC-NFR-001-2. All access to administrative functions must be restricted to users with appropriate privileges.
- SEC-NFR-001-3. User accounts must be subject to a formal provisioning and de-provisioning process.
- SEC-NFR-001-4. Default passwords for any system component must be changed upon first use.
- SEC-NFR-001-5. The system must implement multi-factor authentication (MFA) for administrative accounts.

### 0.4.2 Cryptography (CMCSRS 6.18)

- SEC-NFR-002-1. All data in transit must be encrypted using TLS 1.2 or higher.
- SEC-NFR-002-2. All sensitive data at rest must be encrypted using strong, INSA-approved algorithms.
- SEC-NFR-002-3. Private keys and secrets must never be stored in plaintext.
- SEC-NFR-002-4. Digital signatures must be used for certificate verification as per the Electronic Signature Proclamation.

### 0.4.3 Network Security (CMCSRS 5.7.c)

- SEC-NFR-003-1. A host-based firewall must be active and configured to allow only necessary ports.
- SEC-NFR-003-2. The internal network must be segmented into separate Docker networks.
- SEC-NFR-003-3. All unnecessary services and ports must be disabled.
- SEC-NFR-003-4. Intrusion detection/prevention systems must be deployed.

### 0.4.4 Logging and Monitoring (CMCSRS 6.25)

- SEC-NFR-004-1. All logs must be collected centrally and retained for at least one month.
- SEC-NFR-004-2. Security-relevant events must be logged with timestamps and user identifiers.
- SEC-NFR-004-3. Regular vulnerability scanning must be performed weekly.
- SEC-NFR-004-4. Anomaly detection and alerting must be configured.

### 0.4.5 Incident Response and Business Continuity (CMCSRS 6.26, 6.19)

- SEC-NFR-005-1. An incident response plan must be documented and tested annually.
- SEC-NFR-005-2. A business continuity plan with defined RTO and RPO must be maintained.
- SEC-NFR-005-3. Regular backups must be performed and stored off-site.
- SEC-NFR-005-4. All security incidents must be reported to Ethio-CERT and INSA.

### 0.4.6 Personnel Security (CMCSRS 5.6, 6.37)

- SEC-NFR-006-1. All personnel with access to sensitive systems must undergo security clearance.
- SEC-NFR-006-2. Mandatory annual security awareness training must be provided.
- SEC-NFR-006-3. Job descriptions must include security responsibilities.

### 0.4.7 Compliance and Audit (CMCSRS 6.21, 6.22, 6.23)

- SEC-NFR-007-1. The system must undergo an annual security audit by an INSA-certified auditor.
- SEC-NFR-007-2. All information systems must be accredited before deployment.
- SEC-NFR-007-3. A compliance report must be submitted to INSA annually.

## 0.5 Non-Functional Requirements

### 0.5.1 Performance

- Support at least 10,000 concurrent learners and 500 concurrent educators.
- Page load times below 2 seconds for 95% of requests.
- API response times below 500ms for 95% of requests.
- Video streaming supports at least 720p with adaptive bitrate.

### 0.5.2 Security

- All network communication encrypted with TLS 1.2+.
- Passwords hashed using PBKDF2/BCrypt.
- API authentication using JWT with short expiry.
- RBAC enforced.
- Firewall and intrusion detection in place.



### 0.5.3 Availability

- 99.9% uptime.
- Automatic container restarts.
- Daily off-site backups.

### 0.5.4 Maintainability

- Containerised microservices architecture.
- Quarterly upstream merges.
- Externalised configuration.

### 0.5.5 Scalability

- Horizontal scaling with container replicas.
- Database read replicas supported.

### 0.5.6 Usability

- Responsive interface (modern browsers).
- User-friendly error messages.
- WCAG 2.1 AA accessibility.

## 0.6 Traceability Matrix

| Requirement ID | Test Case ID | Security Reference          |
|----------------|--------------|-----------------------------|
| UFR-OE-001-01  | TC-001       | SEC-NFR-001,<br>SEC-NFR-002 |
| UFR-OE-001-03  | TC-002       | SEC-NFR-001                 |
| UFR-OE-003-01  | TC-003       | SEC-NFR-003,<br>SEC-NFR-004 |
| UFR-OE-003-02  | TC-004       | SEC-NFR-004                 |
| UFR-OE-004-01  | TC-005       | SEC-NFR-002                 |
| UFR-OE-004-03  | TC-006       | SEC-NFR-002                 |
| UFR-OE-005-03  | TC-007       | SEC-NFR-001,<br>SEC-NFR-002 |
| UFR-OE-006-03  | TC-008       | SEC-NFR-001,<br>SEC-NFR-004 |
| UFR-OE-103-01  | TC-009       | SEC-NFR-001                 |
| UFR-OE-106-03  | TC-010       | —                           |
| Continued      |              |                             |

| Requirement ID   | Test Case ID | Security Reference          |
|------------------|--------------|-----------------------------|
| UFR-OE-107-01    | TC-011       | SEC-NFR-002                 |
| UFR-OE-109-01    | TC-012       | SEC-NFR-001,<br>SEC-NFR-003 |
| UFR-OE-110-02    | TC-013       | SEC-NFR-003                 |
| UFR-OE-112-02    | TC-014       | —                           |
| UFR-OE-113-01    | TC-015       | —                           |
| UFR-OE-114-03    | TC-016       | SEC-NFR-006                 |
| SEC-NFR-001..007 | Security doc | CMCSRS,<br>Proc.958/2016    |

## 0.7 Glossary

**CMCSRS** Critical Mass Cyber Security Requirement Standard.

**CMS** Course Management System.

**DFD** Data Flow Diagram.

**INSA** Information Network Security Administration.

**LMS** Learning Management System.

**MFE** Micro-Frontend.

**RBAC** Role-Based Access Control.

**Tutor** Open edX deployment tool.

## 0.8 References

- *Security Architecture & Threat Model – Moove Education Platform*
- CMCSRS v2.0 (2025)
- FDRE National Cybersecurity Policy (2024)
- Computer Crime Proclamation No. 958/2016
- Electronic Signature Proclamation No. 1072/2018

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